# TEMS Academy of Olympiad Math
## Adhoc FE Problem Set

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**Problem 1.** Let $f(x) = x^{67-x^{67-x^{67-\cdots}}}$, where $x \neq 0$. If $f(n) = 64$, then what is the remainder if $n^n$ is divisible by 11?

**Problem 2.** Let

$$f(x) + 2f(-x) = 2x$$

and

$$2f(x) + f(-x) = x^2$$

How many $x$ are there between 1 and 2024 for which both $x$ and $f(x)$ are integers?

**Problem 3.** Let

$$f(x+y) = f(x) + f(y)$$

If $f(1) = 3$, find the value of $f(30)$.

**Problem 4.** Let $f(x,y) = y - x^2$, where $|x| + |y| \leq 21$. Find the difference between the largest and the smallest values of the function. ($x$ and $y$ both are integers)

**Problem 5.** If $f(n) = f(n+1) + f(n-1)$, $f(1) = 5$, $f(2) = 4$, then $f(2024) = ?$

**Problem 6.** Let $f(1) = 2, f(2n) = f(2n-1) + 1$ and $f(2n-1) = f(2n-3) + 2$ (when $n$ is natural number). Find $f(2023)$.

**Problem 7.** If $f(x+3) = f(x+1) + 4x + 12$ and $f(0) = 2$, then what is $f(500)$?

**Problem 8.** Let $S = \{1, 2, 3, \ldots, 12\}$. How many functions $f : S \to S$ are there such that $f(f(x)) = x$ and $f(x) - x$ is divisible by 3?

**Problem 9.** If $f(x) = 3x - 2$, then what is the value of x in the following equation where $f$ is 2020 times iterated?

$$x = f(f(f(\ldots f(x)\ldots)))$$

**Problem 10.** If $f(n) = f(n-1) + 2n - 2$ and $f(1) = 2$, then what is the value of $f(100)$?

**Problem 11.** Let

$$f(3n) = 3f(n)$$
$$f(3n+1) = 3f(n) - 1$$
$$f(3n+2) = 3f(n) - 2$$

If $f(1) = f(2) = 1, f(1009) =$?

**Problem 12.** If $f(x) = 2019$, then what is the value of $f^{1019}(2019)$? **Note:** $f^2(x) = f(f(x))$.

**Problem 13.** Two functions $f(x, y)$ and $g(x)$ are defined on real numbers, such that

$$f(x+y, x-y) = x \cdot g(y) + y \cdot g(x)$$

Given that, $g(x) = 5$ for all values of $x$. Find the value of $f(2019, 0)$.

**Problem 14.** Let $f(1) = 2, f(2) = 5, f(3) = 10$ and

$$f(x) - f(x-3) = 2[f(x-1) - f(x-2)]$$

where $x > 3$. $f(2019) =$?

**Problem 15.** Let $f(4x-3) = 2x + 5$. If $f(2x+1) = ax + b$, $a + b =$?

**Problem 16.** Suppose $f(x)$ is a function. It is given that

$$f(2n-1) = f(2n-2) - 1$$

and

$$f(2n) = 2f(2n-1)$$

If $f(1) = n$, then what is the value of $n$ such that $f(1971) - f(2018) = -1$?

**Problem 17.** If $f(x^2 - 2016x) = xf(x) + 2016$, what is the value of $f(2017)$?

**Problem 18.** Let

$$f(x) = g(x+1)$$
$$f(y) = 2018y + 2016y + \ldots + 2y$$
$$g(x) = h(2x) - x$$

If $h(2018) = a^3$ and $a$ is an integer, $3a =$?

**Problem 19.** For any rational numbers $x, y$, function $f(x)$ is a real number and

$$f(x+y) = f(x)f(y) - f(xy) + 1$$

Again, $f(2017) \neq f(2018)$, $f(\frac{2017}{2018}) = \frac{a}{b}$, where $a, b$ are co-prime. $a + b =$?

**Problem 20.** If $F_0(x) = \frac{1}{1-x}$ and $F_n(x) = F_0(F_{n-1}(x))$. Find $F_{2015}(2016)$.

**Problem 21.** Let

$$f(3m) = \frac{mf(m)}{3}$$
$$f(3m+2) = \frac{(m+2)f(m+2)}{3}$$
$$f(3m+1) = \frac{(m+1)f(m+1)}{3}$$

$f(2016) =$?

**Problem 22.** For all positive integers $x$, $f(f(x)) = 4x + 3$ and for one positive value of integer $k$, $f(5^k) = 5^k \times 2^{k-2} + 2^{k-3}$. $f(2015) =$?

**Problem 23.** Let $f(y) = y$ repeated $y$ times, for example $f(3) = 333, f(5) = 55555$. If $a = f(2001) + f(2002) + \ldots + f(2014) + f(2015)$. What is the remainder upon division of $a$ by 3?

**Problem 24.** $f(n) =$ sum of the squares of digits of $n$. If $f_k(n) = f(f_{k-1}(n))$, $f_{2014}(3) =$?

**Problem 25.** For all positive integers $x, y$, $f(x) \geq 0$ and $f(xy) = f(x) + f(y)$. If the digit at the one's of $x$ is 6, then $f(x) = 0$. If $f(1920) = 420$, then $f(2015) =$?

**Problem 26.** If $x, a, b$ are positive integers, $f(x)$ is positive integer too, and if $a > b$, then $f(a) > f(b)$. Again, $f(f(x)) = x^2 + 2$. Find the value of $f(6)$.

**Problem 27.** A function $f : \mathbb{R} \to \mathbb{R}$ is defined in such a way that $f(x+y) = f(x) \cdot f(y)$, and for $a \in \mathbb{N}$, $\sum_{k=1}^{n} f(a+k) = 16(2^n - 1)$. If $f(1) = 2$, then what is the value of $a$?

**Problem 28.** Given that, $F(x, y) = F(xy, \frac{x}{y})$ where $y \neq 0$. If $F(x^4, y^4) + F(x^2, y^2) = 16$, then $F(x, y) =$?

**Problem 29.** If $f(x, y) = x + y \times f(y, x)$, then $f(2, 1) =$?

**Problem 30.** A function $f : \mathbb{R} \to \mathbb{R}$ satisfies $f(1) = 1$ and for any $x \in \mathbb{R}$, $f(x+7) \geq f(x) + 7$ and $f(x+1) \leq f(x) + 1$. $f(2013) =$?

**Problem 31.** Let $f : \mathbb{R} - \{0\} \to \mathbb{R}$ be a function such that $f(xy) = \frac{f(x)}{y}$. If $f(2012) = 1$, then $f(2013) =$?

**Problem 32.** The function $f(x)$ satisfies the equation

$$(x-2)f(x) + f(\frac{1}{x}) = 1$$

Find the value of $f(3)$.

**Problem 33.** If $f(x) = a + bx$ and $f(f(f(x))) = 27x + 52$, and if $a$ and $b$ are real numbers, then $a - b =$?

**Problem 34.** What is the domain of the function $f(x) = \log_{|x|} \sqrt{x^2}$?

**Problem 35.** For an injective function $f : \mathbb{R} \to \mathbb{R}$,

$$f(x + f(y)) = 2012 + f(x + y)$$

$f(2013) =$?